Chapter 1: Why Decision-Making Fails

We make decisions every day.

What to choose, how to act, what to prioritize. These judgments are not determined by ability or knowledge alone.

Consider the statement "I want to travel alone." In most cases, this is received as a request for travel advice. But the person saying it might be a child using a parent's smartphone. It might be an elderly person with dementia. It might be someone carrying serious debt. When the premise differs, the appropriate response changes fundamentally.

The same problem appears in conflict situations. "I demand compensation and damages" is easily interpreted as a financial claim. In reality, the person may simply want their lost data restored.

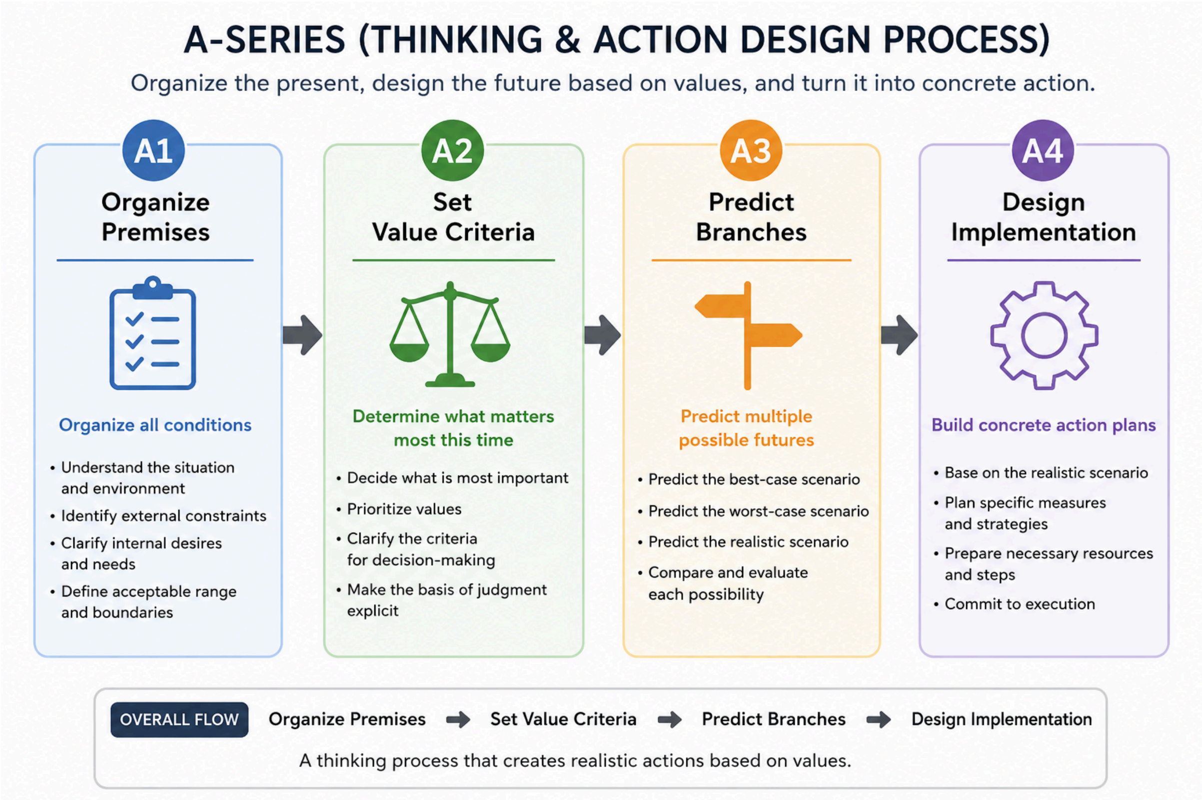
This problem is not limited to human dialogue. It may also be observed in many dialogue systems, including current LLMs.

And this premise mismatch does not stop at conversational misunderstanding. When judgment proceeds without a shared premise, every subsequent choice, priority, and action plan accumulates on a misaligned foundation.

Most failures in decision-making are not failures of ability. They may be structural failures — judgment beginning before premises have been confirmed.

The A-Series is a model that organizes this structure into four stages of the decision-making process.

Chapter 2: What Is the A-Series?



The A-Series is the thought and action design module within COS (Cognitive Operating System).

Its role is not to produce correct conclusions. It is to organize the process by which decisions are formed and to decompose that process into observable components.

Decision-making is often perceived as a single act. In reality, however, multiple distinct processes exist before a judgment is reached.

The A-Series organizes this process into four functions.

・A1: Premise Clarification
・A2: Value Criteria
・A3: Divergence Prediction
・A4: Implementation Design

A1 clarifies the premises that form the foundation of judgment. A2 makes explicit the value criteria that determine what is prioritized. A3 predicts the multiple possible outcomes that arise from a given choice. A4 converts the selected direction into an actionable form.

These functions do not necessarily execute in a single linear sequence. In actual decision-making, they may interact and cycle back on one another.

However, by decomposing the process into these four functions, it becomes easier to identify where judgment became distorted and where room for improvement exists.

The purpose of the A-Series is not to guarantee correct conclusions. It is to make the decision-making process observable as a structure.

Chapter 3: A1 — Premise Clarification

A1 is the function of clarifying "what is treated as premise" before judgment begins.

A1 is placed first because, without clear premises, the value criteria and divergence predictions that follow cannot be stable. No matter how refined the judgment, if the foundation is unstable, deviation begins at the starting point.

In A1, premises such as situational and environmental conditions, external constraints, internal needs, and tolerance boundaries are organized to make the starting point of judgment observable. These are representative observation items referenced when confirming premises — they do not constitute a strict classification system.

The purpose of A1 is not to derive correct premises. It is to make explicit what has been placed as the foundation of judgment.

Chapter 4: A2 — Value Criteria

A2 is the function of clarifying the evaluative axis that determines what is prioritized, based on the premises organized in A1.

Even for the same person, the prioritized evaluative axis may change when the premise changes. For example, consider someone who had no plans for the weekend and decided to go out, only to find it was raining outside. In this case, the value criteria may shift from prioritizing a change of scenery to prioritizing comfort and avoiding burden. A single change in premise can alter the same person's judgment.

What A2 handles is not values themselves. It is the evaluative axis used to determine what is prioritized in decision-making. Values are the beliefs, preferences, and priority tendencies held by a person or organization, and they are not always made explicit. From these, A2 makes the order of priorities explicit in a form that can be used for judgment under the current premise conditions.

A2 does not exist independently of A1. Value criteria are not attributes of personality — they may change through interaction with the premise conditions organized in A1.

The purpose of A2 is not to analyze values. It is to make explicit, in an observable form, what is to be prioritized under the premises organized in A1.

Chapter 5: A3 — Divergence Prediction

A3 is the function of predicting multiple possible outcomes of a choice, based on the value criteria clarified in A2.

In decision-making, judgment typically proceeds with only the most likely outcome in mind. In A3, however, no single outcome is converged upon. Multiple futures are held in parallel across three axes: best case, worst case, and realistic case.

The purpose is not to prepare for the worst. By holding multiple outcomes simultaneously, A4's action design avoids becoming a bet on a single scenario.

The three axes of A3 can be organized as follows. The best-case outcome is the scenario in which the value criteria defined in A2 are most fully satisfied. The worst-case outcome is the scenario in which those value criteria are not merely unmet, but progress in the opposite direction of the intended purpose. The realistic outcome is the scenario estimated to be most likely based on information currently available. These three outcomes are not fixed. If what is prioritized in A2 changes, the evaluation of best and worst may change even in the same situation.

Consider a situation where someone is invited to a party. Suppose A2 has clarified the value criterion of "prioritizing an enjoyable time." The best-case outcome is spending time with compatible people and having a highly satisfying experience. The realistic outcome is the state estimated to be most likely given the participants, atmosphere, and past experience. The worst-case outcome is not merely failing to enjoy oneself, but experiencing interpersonal conflict that leads to exhaustion or damaged relationships. A3 does not select one of these as the correct answer. By holding multiple outcomes simultaneously, it creates the conditions for A4 to design actions that can respond to multiple scenarios.

The purpose of A3 is not to predict the correct future. It is to create the conditions for A4's action design to accommodate multiple possible outcomes.

Chapter 6: A4 — Implementation Design

A4 is the function of designing executable actions based on the multiple outcomes retained in A3.

The role of A4 is not to select one future and commit to it entirely. It designs actions that aim toward the best-case outcome, avoid the worst-case outcome, and remain capable of responding even if the worst case occurs. By simultaneously holding the design for achieving the best case, the design for avoiding the worst case, and the design for responding if the worst case occurs, A4 creates a state where outcomes above the realistic range can be pursued while remaining capable of responding if the worst case arrives.

Consider the party example. Suppose A2 has clarified the value criterion of "prioritizing an enjoyable time," and A3 has retained the best, realistic, and worst-case outcomes. At this point, A4 designs concrete actions aimed at achieving the best case while avoiding the worst. One output that naturally emerges from this process is the question: "Who's coming?" This is an action designed as a result of passing through A1 through A3 — an example of the A-Series process appearing as a concrete output.

The purpose of A4 is not to create a perfect plan. It is to prepare, in executable form, a state that can respond to multiple possible outcomes.

Chapter 7: Example

The previous chapters explained A1 through A4 individually. Chapter 7 uses the same party example to show how A1 through A4 function as a single continuous process.

Situation
Someone has been invited to a party over the weekend. The judgment to be made is whether to attend.

A1: Premise Clarification
First, the premises that form the foundation of judgment are organized. Who will be attending is not yet known. The location and time have been confirmed. There is work the following day. Physical condition is not a concern.

What was clarified at this stage was the set of premise conditions affecting the judgment. Among these, the fact that the attendees are unknown was observed as a factor that would significantly influence subsequent judgment.

A2: Value Criteria
Next, what is to be prioritized is determined. In this case, the value criterion of "prioritizing an enjoyable time" was made explicit.

What was established at this stage was the evaluative axis for judgment. The next question becomes how the participants and atmosphere will affect this value criterion.

A3: Divergence Prediction
Based on the value criteria from A2, multiple outcomes are retained. The best-case outcome is spending time with compatible people and having a highly satisfying experience. The realistic outcome is the state estimated to be most likely given the participants and past

experience. The worst-case outcome is experiencing interpersonal conflict that leads to exhaustion or damaged relationships.

What was retained at this stage were three outcomes. No single outcome is converged upon.

A4: Implementation Design
Based on the three outcomes retained in A3, actions are designed. The aim is to pursue the best case, avoid the worst case, and remain capable of responding if the worst case occurs.

The output that emerges from this process is the question: "Who's coming?" This is an act of information gathering designed to draw the best-case outcome closer while avoiding the worst. Securing a means of getting home in advance is also included in the design as preparation for the worst-case outcome.

Looking across A1 through A4, it becomes clear that the single question "Who's coming?" was generated through four stages of processing: premise clarification, value criteria, divergence prediction, and implementation design. The A-Series is a model that makes this processing observable as a structure.

Chapter 8: Connection to the COS Series

The A-Series is a model that makes the decision-making process observable as a structure. However, the A-Series does not function in isolation.

In actual cognitive processing, decision-making operates within an interplay of other processing systems. Processes related to interpretation, communication, and observation exist in parallel with, or before and after, the A-Series. COS is a framework that integrates these processing systems, and the A-Series constitutes one part of it.

There are phenomena that cannot be fully explained within the observational scope of the A-Series alone.

For example, in dialogue, cases have been observed in which premise clarification does not sufficiently activate even in situations where it appears necessary. This may not be a problem with the A-Series itself. There is a possibility that the processing occurring before the A-Series engages — namely, how the other party's statement was interpreted — influences the starting point of judgment.

Before decision-making begins, there exists a process of determining how to interpret what the other party has said. Within COS, this domain is handled by the B-Series, the interpretation system.

The B-Series will be addressed in the next paper.